

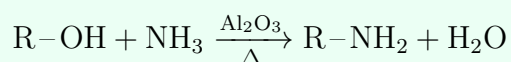
Chapter 9: Organic Compounds Containing Nitrogen

Named Reactions Sheet + Tips & Tricks

Formula Sheet

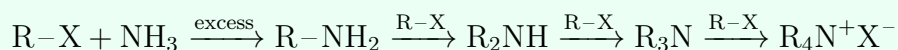
I. Preparation of Amines

From Alcohols



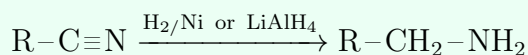
Vapours passed over heated alumina; mixture of 1°/2°/3° amines usually formed.

From Alkyl Halides – Ammonolysis



Excess NH_3 favours the primary amine; excess R-X drives further alkylation up to the quaternary ammonium salt.

From Nitriles – Reduction



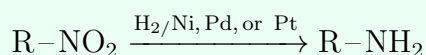
Gives a primary amine with one more carbon than the parent alkyl halide.

From Amides – Hofmann Bromamide Degradation



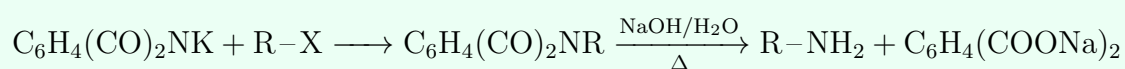
Converts an amide to a primary amine with *one less* carbon (the carbonyl carbon is lost as carbonate); proceeds via an isocyanate intermediate.

From Nitro Compounds – Reduction



Also achieved with Sn/HCl or Fe/HCl .

Gabriel Phthalimide Synthesis

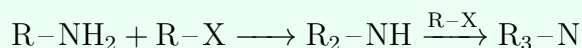


Gives a pure primary amine only (aryl halides do not work since they don't undergo nucleophilic

substitution under these conditions).

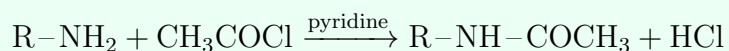
II. Chemical Properties of Amines

Alkylation



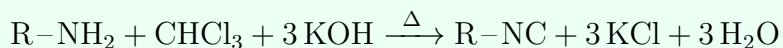
Successive alkylation; excess alkyl halide drives the sequence to the quaternary ammonium salt.

Acylation



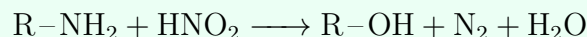
Forms a substituted amide; pyridine absorbs the HCl by-product.

Carbylamine Reaction (Isocyanide Test)



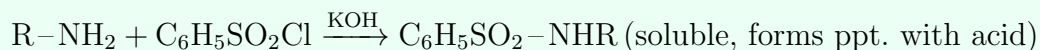
Positive only for 1° amines (aliphatic or aromatic); gives a foul-smelling isocyanide – diagnostic test for primary amines.

Reaction with Nitrous Acid – Primary Aliphatic Amine



Brisk effervescence of N₂ gas; quantitative – basis of the van Slyke method for estimating primary amino groups.

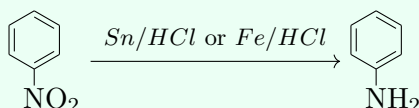
Hinsberg's Test – Distinguishing 1°/2°/3° Amines



1° amine gives a KOH-soluble sulfonamide; 2° amine gives a KOH-insoluble sulfonamide; 3° amine does not react (Hinsberg's reagent = benzenesulfonyl chloride).

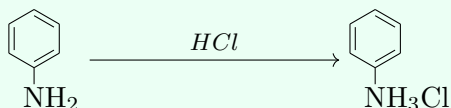
III. Aniline – Preparation and Reactions

Preparation – Reduction of Nitrobenzene



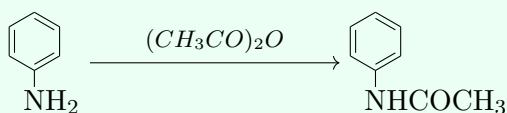
Standard industrial/lab route to aniline; the metal-acid salt formed is then treated with NaOH to liberate free aniline.

Reaction with HCl



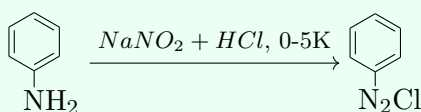
Forms anilinium chloride (a salt) – aniline is a weaker base than ammonia/alkyl amines due to delocalisation of the lone pair into the ring.

Acetylation



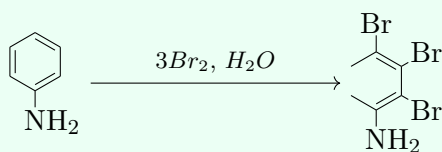
Forms acetanilide; used to moderate aniline's reactivity before further ring substitution.

Diazotisation



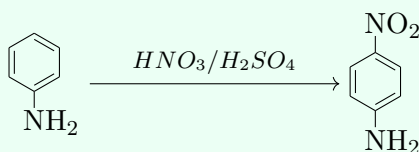
Forms benzenediazonium chloride – gateway to Sandmeyer, Gattermann, Balz-Schiemann, and coupling reactions (see Section V).

Electrophilic Substitution – Bromination



-NH₂ is strongly activating and *o,p*-directing; reaction proceeds readily even without a catalyst, giving the 2,4,6-tribromo product directly.

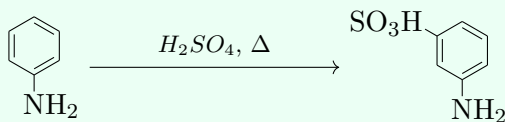
Electrophilic Substitution – Nitration



Aniline is first protonated/acetylated to moderate the ring's reactivity and avoid oxidation by conc.

HNO_3 at 273 K; para is the major product.

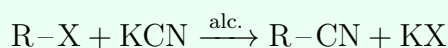
Electrophilic Substitution – Sulphonation



Forms sulphanilic acid, mainly via the anilinium hydrogen sulphate intermediate rearranging on heating.

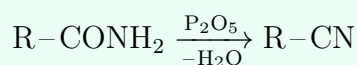
IV. Cyanides and Isocyanides

Cyanide from Alkyl Halide



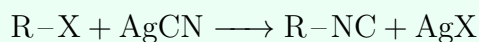
Nucleophile attacks via carbon (KCN is ionic) – gives the alkyl cyanide/nitrile.

Cyanide from Amide – Dehydration



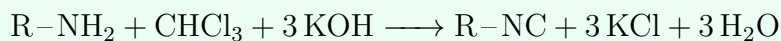
Dehydration of an unsubstituted amide gives the nitrile.

Isocyanide from Alkyl Halide



Nucleophile attacks via nitrogen (AgCN is covalent, follows Fajan's rule) – gives the isocyanide.

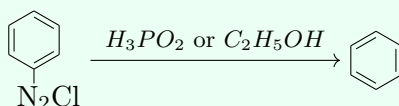
Isocyanide from Primary Amine (Carbylamine Reaction)



Same reaction used as the diagnostic test for 1° amines.

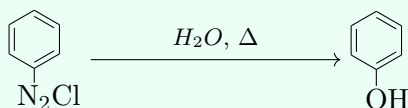
V. Diazonium Salts – Reactions

Replacement by -H (Reduction)



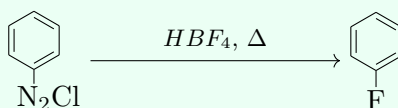
Hypophosphorous acid or ethanol reduces the diazonium group to H, effectively removing the -NH_2 group used to direct earlier substitution.

Replacement by -OH



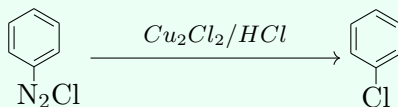
Warm hydrolysis gives phenol.

Replacement by -F (Balz-Schiemann)



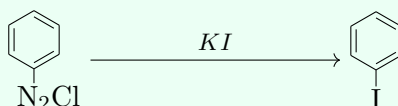
Standard route to fluorobenzene.

Replacement by -Cl/-Br (Sandmeyer)



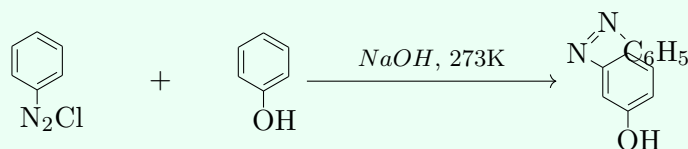
Cuprous halide catalyses the substitution (also works with $\text{Cu}_2\text{Br}_2/\text{HBr}$ for the bromo product).

Replacement by -I



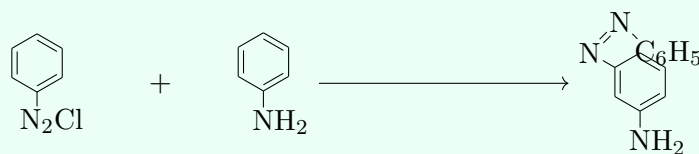
Unlike Cl/Br, iodide substitution proceeds directly without a catalyst.

Coupling Reaction with Phenol



Electrophilic diazonium ion attacks the activated phenol ring, forming an azo dye (p-hydroxyazobenzene); coupling occurs at the para position (or ortho if para is blocked).

Coupling Reaction with Aniline



Forms p-aminoazobenzene.

Tips, Tricks & Hacks

- **Basicity order in water:** for aliphatic amines, $2^\circ > 1^\circ > 3^\circ > NH_3$ (steric hindrance and solvation effects push 3° below $1^\circ/2^\circ$ despite the extra alkyl groups); aniline is far weaker than any aliphatic amine since the lone pair delocalises into the ring.
- **Carbylamine test** is positive *only* for primary amines (1°) – both aliphatic and aromatic. This is the single most reliable "which amine is this" test question.
- **Hinsberg's test** distinguishes all three classes in one reagent: soluble sulfonamide = 1° , insoluble sulfonamide = 2° , no reaction = 3° .
- **Hofmann degradation shortens the chain by one carbon;** Gabriel synthesis and nitrile reduction *keep or extend* the chain – keep these three "amine-forming" routes straight by tracking carbon count.
- **Diazonium salt chemistry is only stable at 0–5 K (ice-cold);** above this temperature it decomposes, especially releasing N_2 and forming phenol on warming with water – exploited deliberately in the -OH replacement reaction.
- **Coupling reactions** (azo dye formation) work only with strongly activated rings (phenols, aromatic amines) and occur predominantly at the para position.
- $-NH_2/-NHR/-NR_2$ are the *strongest* activating, *o,p*-directing groups on a benzene ring – stronger than $-OH/-OR$ – which is why aniline reacts with bromine water without any catalyst at all.